tion of two different algorithms attached to these two fields independently: `title` is associated with the `BM25` algorithm, while `text` is set with `boolean`.

Each similarity function has additional parameters, and we can alter them to reflect precise search results. For example, although the BM25 function is set by default with the optimal parameters, we can easily modify the function using the index settings API. We can change two parameters in BM25 if we need to: `k1` and `b`, described in table 3.9.

Table 3.9 Available BM25 similarity function parameters

Property	Default value	Description
k1	1.2	Nonlinear term frequency saturation variable
b	0.75	TF normalization factor based on the document's length

Let's look at an example. Figure 3.21 shows an index with a custom similarity function, where the core BM25 function is amended with our own settings for `k1` and `b`.

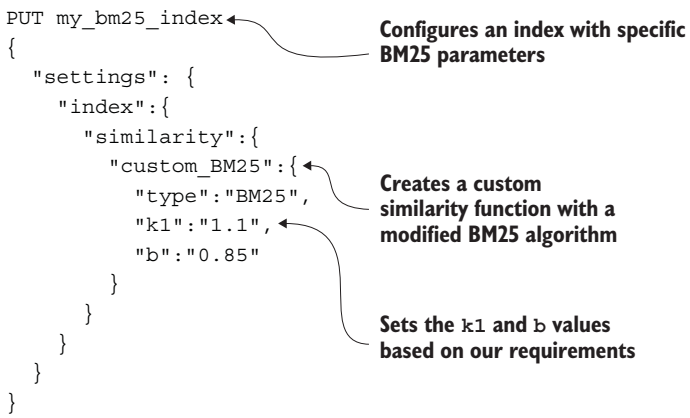


Figure 3.21 Setting custom parameters on the BM25 similarity function

Here, we are creating a custom similarity type—a tweaked version of BM25, which can be reused elsewhere. (It's more like a data type function, predefined and ready to be attached to attributes.) Once this similarity function is created, we can use it when setting up a field, as shown in figure 3.22.